

## Contact

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GitHub

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Online Resume / Portfolio

[mahdirahimi1994.github.io/Resume/](https://mahdirahimi1994.github.io/Resume/)

## About Me

I am a Web Developer with nearly three years of experience building research-focused web applications using React.js, Node.js, Express.js, SQL and Docker. I have worked on API-driven tools that connect passport and genotype data from platforms such as Genesys-PGR, Gigwa and Germinate. My strengths include front-end development, backend services, deployment support, documentation and working closely with developers, researchers and data managers. I have also completed AI-focused training covering Large Language Model APIs, prompt design, structured outputs and AI-assisted software development.

## Education

**Master of Information Technology** (2020-2023)  
La Trobe University - Melbourne, VIC

- 89.7 GPA

## Awards

- La Trobe University Student Excellence Recognition
- La Trobe University International Merit Scholarship
- Recognised by the Australian Computer Society for successful skills assessment as a Web Developer.

## Certificates

- Introduction to AI Engineering
- Learn to Code With AI
- HTML Essential Training
- CSS Essential Training
- User Experience for Web Design
- UX Foundations: Accessibility
- JavaScript Essential Training
- React.js Essential Training
- Git Essential Training
- Docker Essential Training
- Node.js Essential Training
- Certificates available upon request

## Technical Skills

- Frontend Development:**  
HTML, CSS, JavaScript, TypeScript, React.js, Redux, Vite, responsive UI development, reusable component design, CSS Modules
- Backend Development:**  
Node.js, Express.js, REST API development, MySQL, Sequelize, Dolt, backend validation and file upload handling. Experienced in building backend endpoints that connect to Genesys-PGR, Gigwa, Germinate and BRAPI services; retrieve passport, accession, sample, dataset and genotype information; combine API responses; and prepare the returned data for display in React-based interfaces.
- Databases and Data Formats:**  
MySQL, Sequelize, Dolt and MongoDB. Experienced working with VCF-based genotype data, Gigwa/Germinate database concepts, passport records from Genesys-PGR, accession identifiers, genotype/sample IDs, datasets, studies/projects and related metadata used in genebank and genomic data workflows.
- Genebank and Genomic Data Knowledge:**  
Familiar with the relationship between passport data and genotype data, including how accession records from Genesys-PGR can be linked with genotype/sample IDs, VCF-based genotype datasets, studies/projects and marker or position-based genotype queries in platforms such as Gigwa and Germinate.
- Deployment and Server Management:**  
Docker, Docker Compose, AWS EC2, Nginx, SSL/HTTPS using Certbot, Route 53, Security Groups, Linux-based server configuration
- Authentication and Security:**  
JWT, OpenID Connect/OAuth 2.0 workflows, react-oidc-context, token-based authentication, secure third-party server communication
- Version Control and Collaboration:**  
Git, GitHub, GitLab, branch management, collaborative software-development workflows, documentation and issue tracking
- Testing and Tools:**  
Jest, unit and integration testing, Figma, technical documentation
- AI-Assisted Development:**  
Familiar with Large Language Model APIs, AI API integration concepts, model selection, prompt design, structured outputs and AI-assisted project development workflows. Completed AI-assisted and web development projects like Polly-Glot

# Work History

## Full-time web-developer

AgriBio Centre for AgriBioScience, Agriculture Victoria  
November 2023 – Now

### Key Responsibilities:

- Developed, maintained and enhanced Genolink, a middleware web application that supports integration between genotype databases and Genesys-PGR, a genebank passport data repository.
- Contributed to the public Genolink codebase: [github.com/plantinformatics/genolink](https://github.com/plantinformatics/genolink)
- Designed and implemented React.js front-end features and Node.js/Express.js backend services to support searching, filtering, data retrieval and presentation of passport and genotype-related data.
- Built backend endpoints that connect to Genesys-PGR, Gigwa, Germinate and BRAPI services to retrieve passport records, accession details, genotype/sample IDs, dataset/study information and genotype-related results.
- Combined API responses and prepared structured outputs for display in React-based interfaces.
- Improved traceability between accession records, genotype datasets, samples and project/study sources.
- Supported deployment, configuration and maintenance of public-facing applications using Docker, Docker Compose, Nginx, SSL/HTTPS and AWS EC2-based infrastructure.
- Contributed to authentication and authorised data access workflows using token-based authentication and secure communication with third-party servers.
- Designed and developed SPOT, a web application to centralise data management between Agriculture Victoria Research and InterGrain, reducing reliance on Excel file exchange with varying templates.
- Implemented SPOT features including backend validation, file upload handling, database rules, data tracking workflows and user-error reduction.
- Developed SPOT within an internal GitLab-based workflow, using version control and collaborative development practices to manage application changes and releases.
- Improved application stability, usability and performance through user feedback, debugging, code refactoring and regular feature updates.
- Continued developing knowledge of AI-assisted web development and Large Language Model API workflows, including how AI-based search and natural-language interfaces could support more intuitive access to complex research and genebank data.
- Prepared technical documentation and communicated development progress with software developers, researchers, data managers and scientific stakeholders.

### Achievements:

- Improved Genolink functionality and usability by delivering regular updates based on user feedback, supporting better access to integrated passport and genotype data.
- Enhanced data traceability between accessions, passport records, genotype datasets and sample/project sources, supporting more transparent research data discovery.
- Contributed to the evolution of Genolink as a research data platform by improving API integration, front-end workflows, data presentation and deployment reliability.
- Designed and delivered SPOT as a structured data management application using React.js, Node.js/Express.js, Dolt, Docker and AWS EC2.
- Reduced manual and error-prone data exchange processes in SPOT by replacing inconsistent Excel-based workflows with database-driven validation, backend checks and clearer responsibility tracking.
- Supported secure and reliable application deployment through Docker-based workflows, SSL configuration, Nginx server management and AWS infrastructure setup.

## Casual web-developer

AgriBio Centre for AgriBioScience, Agriculture Victoria  
June 2023 – November 2023

### Key Responsibilities:

- Investigated genomic data platforms including Gigwa, Germinate and Google Genomics to understand data access, API behaviour and genotype data workflows.
- Set up and configured a Gigwa server on AWS EC2 to support genotype data access, testing and API investigation.
- Investigated Gigwa API endpoints responsible for retrieving and filtering genotype data from VCF-based datasets.
- Created the initial version of Genolink to interface with Genesys-PGR and genomic platforms such as Germinate and Gigwa through API-based workflows.
- Built backend endpoints to retrieve passport data from Genesys-PGR and genotype-related data from Gigwa and Germinate.
- Developed React.js front-end components to visualise integrated passport and genotype search results.
- Worked with external platform support teams to understand API access, authorisation requirements and integration behaviour.

### Achievements:

- Successfully retrieved and filtered large-scale genotype data through Gigwa API endpoints.
- Developed the initial Genolink workflow to integrate Genesys-PGR passport data with genotype data from genomic platforms.
- Enabled smoother data exchange by creating backend API endpoints that connected multiple data sources.
- Set up a Gigwa server on AWS EC2, supporting secure and efficient research data processing and testing.

## **Backend Intern**

AgriBio Centre for AgriBio science, Agriculture Victoria  
January 2023 - June 2023

### **Key Responsibilities:**

- Set up and configured the Germinate application by working with client-side and server-side source code from the Germinate GitHub repository.
- Built the front-end for a production environment and deployed the server-side application on Tomcat using Gradle on AWS.
- Used Certbot to generate SSL/HTTPS certificates for secure communication with the deployed Germinate application.
- Created Dockerfiles for MySQL and Tomcat server containers to support deployment and environment management.
- Investigated the Germinate backend to understand where genotype data was stored and how different data types were structured in the database.
- Identified backend API endpoints responsible for importing and exporting genotype data.
- Added backend endpoints to retrieve datasets, retrieve samples within datasets and retrieve genotype values based on sample lists, genomic position intervals or marker IDs.

### **Achievements:**

- Improved understanding of genomic data storage and API-based genotype data retrieval workflows.
- Supported deployment and secure configuration of Germinate using AWS, Tomcat, Docker, MySQL and SSL/HTTPS.
- Contributed backend functionality to improve access to dataset, sample and genotype information.